Decision Tree Variants Impleme | Untitled0.ipynb - Colab | Breast Cancer Dataset | Ask Gemini

colab.research.google.com/drive/1xZ5MWRVBnGwJ3LmGOqFPZgeYy2MdJ7ya#scrollTo=GGDOmHTSuG4Q

Untitled0.ipynb

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About this dataset

Breast Cancer Data...

yasserh • 666 upvotes • 4 y ago

Description: Breast cancer is the most common cancer amongst women in the world. It accounts for 25% of all cancer cases, and affected over 2.1 Million people i... Read more

Code snippet

```
import kagglehub
path = kagglehub.dataset_
```

Dataset

- breast-cancer-data...
  - breast-cancer.csv

Summary

- 1 file

[21] 5s

```
!pip install scikit-learn matplotlib graphviz
!apt-get install graphviz -y
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: graphviz in /usr/local/lib/python3.12/dist-packages (0.21)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
graphviz is already the newest version (2.42.2-6ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

[22] 3s

```
import kagglehub
path = kagglehub.dataset_download("yasserh/breast-cancer-dataset")
```

```
Using Colab cache for faster access to the 'breast-cancer-dataset' dataset.
```

[23]

Variables Terminal

20:26 Python 3

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Code snippet

```
import kagglehub
path = kagglehub.dataset_download("yasserh/breast-canc
```

Dataset

- breast-cancer-dataset
  - breast-cancer.csv

Summary

- 1 file
- 32 columns

[25] 0s

```
# 3. Random Forest
rf = RandomForestClassifier(n_estimators=10, random_state=42)
rf.fit(X_train, y_train)

# 4. Extra Trees
et = ExtraTreesClassifier(n_estimators=10, random_state=42)
et.fit(X_train, y_train)

# 5. Gradient Boosting
gb = GradientBoostingClassifier(n_estimators=10, random_state=42)
gb.fit(X_train, y_train)

# 6. AdaBoost
ada = AdaBoostClassifier(n_estimators=10, random_state=42)
ada.fit(X_train, y_train)
```

```
AdaBoostClassifier
AdaBoostClassifier(n_estimators=10, random_state=42)
```

[26] 0s

```
def print_tree(tree, depth=0):
    if not isinstance(tree, dict):
        print("  " * depth + f"Leaf: {tree}")
        return

    print("  " * depth + f"[X{tree['feature']} <= {tree['value']}]")
    print_tree(tree['left'], depth + 1)
    print_tree(tree['right'], depth + 1)
```

Variables Terminal

20:26 Python 3

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```

Dataset

- breast-cancer-dataset
  - breast-cancer.csv

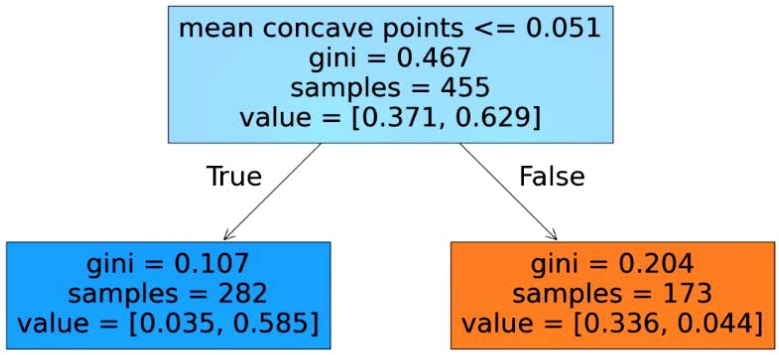
Summary

- 1 file
- 32 columns

[34]

```
plt.figure(figsize=(20,10))
plot_tree(ada.estimators_[0],
          feature_names=data.feature_names,
          filled=True)
plt.title("AdaBoost - Tree 1")
plt.show()
```

AdaBoost - Tree 1



Variables Terminal

20:26 Python 3

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```

Dataset

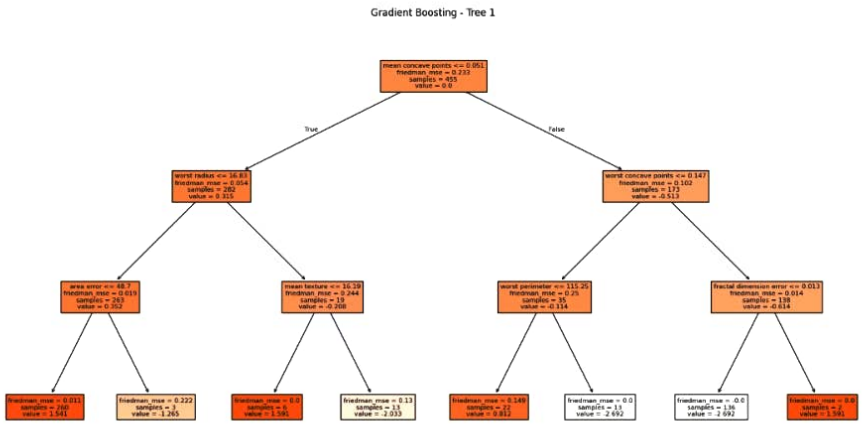
- breast-cancer-dataset
  - breast-cancer.csv

Summary

- 1 file
- 32 columns

[33]

```
plt.figure(figsize=(20,10))
plot_tree(gb.estimators_[0, 0],
          feature_names=data.feature_names,
          filled=True)
plt.title("Gradient Boosting - Tree 1")
plt.show()
```



Variables Terminal

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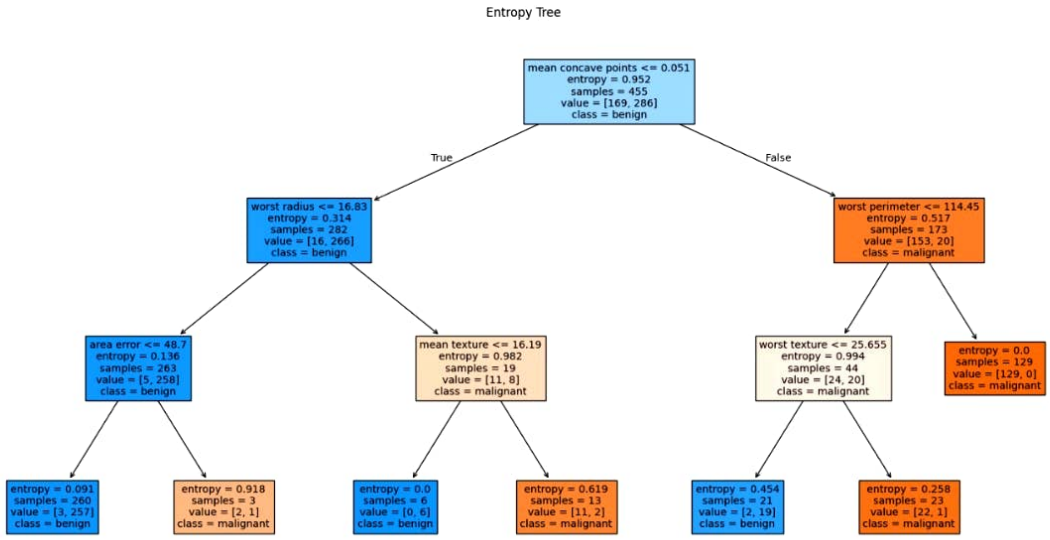
[29]
0s

ost common cancer amongst
25% of all cancer cases, and
Read more

```
unload("yasserh/breast-canc
```

```
plt.figure(figsize=(20,10))
plot_tree(entropy_tree, feature_names=data.feature_names,
          class_names=data.target_names, filled=True)
plt.title("Entropy Tree")
plt.show()
```



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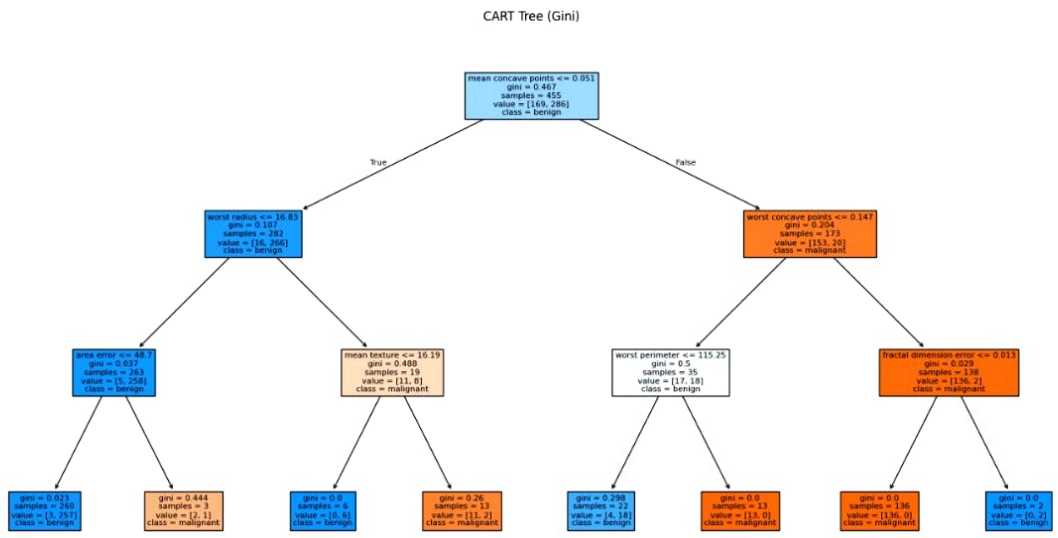
colab.research.google.com/drive/1xZ5MWRVBnGwJ3LmGOqFPZgeYy2MdJ7ya#scrollTo=GGDOmHTSuG4Q

ost common cancer amongst
25% of all cancer cases, and
Read more

```
unload("yasserh/breast-canc
```

[27]
0s

```
plt.figure(figsize=(20,10))
plot_tree(cart, feature_names=data.feature_names,
          class_names=data.target_names, filled=True)
plt.title("CART Tree (Gini)")
plt.show()
```



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Code snippet

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import kagglehub
path = kagglehub.dataset_download("yasserh/breast-canc
```

Dataset

- breast-cancer-dataset
  - breast-cancer.csv

Summary

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```
    [X7 <= 0.05074]
      [X20 <= 16.77]
        [X10 <= 0.6061]
          [X21 <= 30.04]
            Leaf: 1
            [X20 <= 14.37]
              Leaf: 1
              [X0 <= 13.44]
                [X1 <= 26.86]
                  Leaf: 0
                  Leaf: 1
                Leaf: 1
          [X4 <= 0.08749]
            Leaf: 1
            Leaf: 0
        [X1 <= 15.7]
          Leaf: 1
          [X17 <= 0.009921]
            Leaf: 0
            Leaf: 1
      [X22 <= 114.3]
        [X21 <= 25.47]
          [X27 <= 0.1571]
            Leaf: 1
            Leaf: 0
          [X12 <= 1.513]
            [X0 <= 13.17]
              Leaf: 0
              Leaf: 1
            Leaf: 0
        Leaf: 0
```

Variables Terminal

20:26 Python 3

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colab.research.google.com/drive/1xZ5MWRVBnGwJ3LmGOqFPZgeYy2MdJ7ya#scrollTo=GGDOmHTSuG4Q

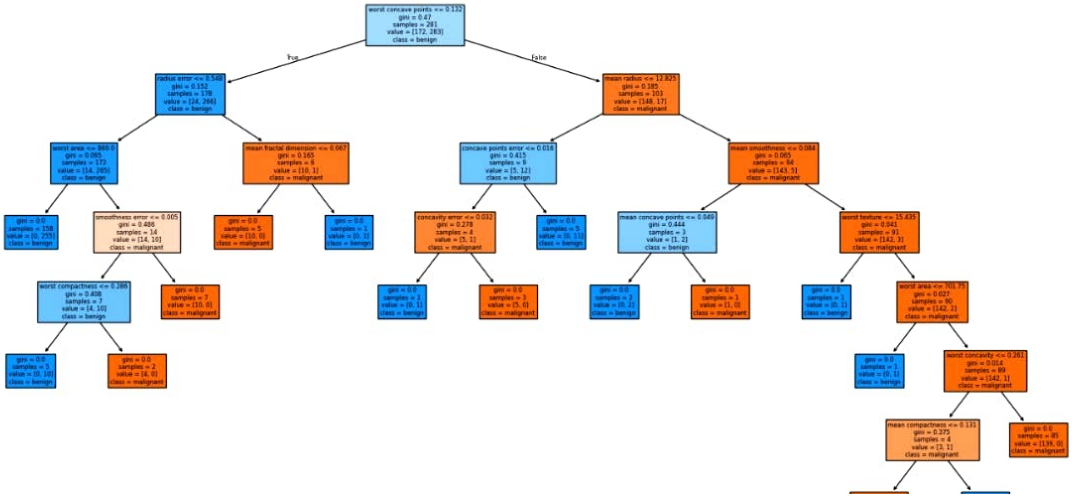
ost common cancer amongst
25% of all cancer cases, and
Read more

```
unload("yasserh/breast-canc
```

[31]
1s

```
plt.figure(figsize=(20,10))
plot_tree(rf.estimators_[0],
          feature_names=data.feature_names,
          class_names=data.target_names,
          filled=True)
plt.title("Random Forest - Tree 1")
plt.show()
```

Random Forest - Tree 1



Decision Tree Variants Impleme | Untitled0.ipynb - Colab | Breast Cancer Dataset | Ask Gemini

colab.research.google.com/drive/1xZ5MWRVBnGwJ3LmGOqFPZgeYy2MdJ7ya#scrollTo=GGDOmHTSuG4Q

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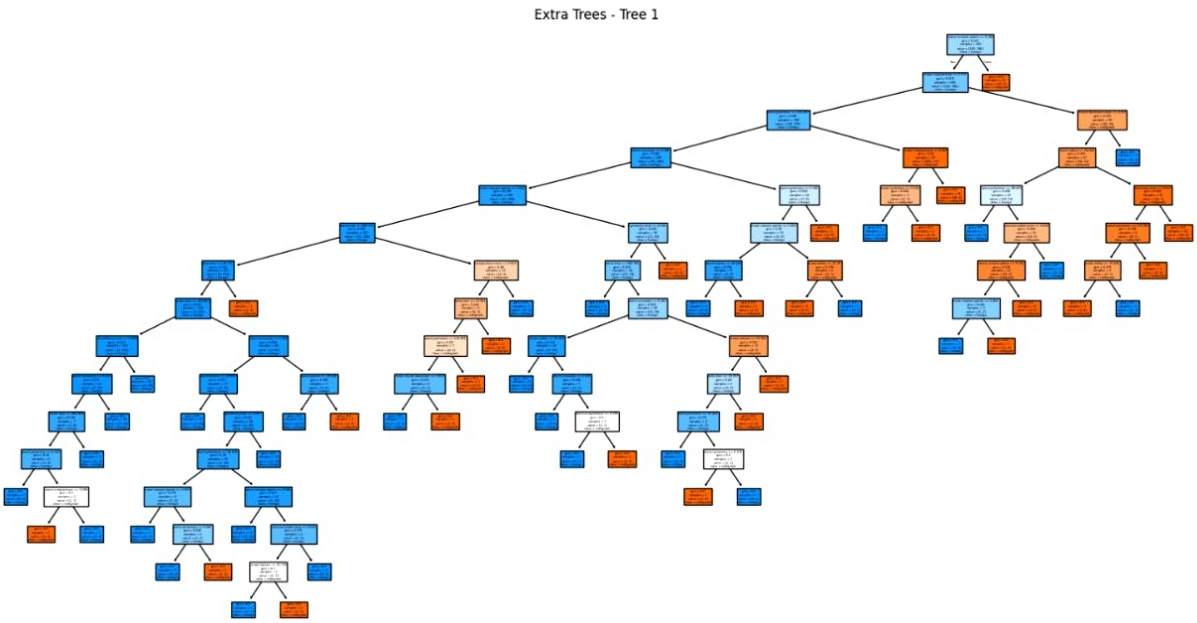
cancer cases, and

asserh/breast-canc

[32]
3s

```
plt.show()
```



Welcome | dt.py

dt.py > ...

```python
models = {
    "CART": cart,
    "Entropy Tree": entropy_tree,
    "Random Forest": rf,
    "Extra Trees": et,
    "Gradient Boosting": gb,
    "AdaBoost": ada
}

for name, model in models.items():
    preds = model.predict(X_test)
    print(f"{name} Accuracy:", accuracy_score(y_test, preds))

    plt.figure(figsize=(20,10))
plot_tree(cart, feature_names=data.feature_names, class_names=data.target_names, filled=True)
plt.title("CART Tree")
plt.show()

plt.figure(figsize=(20,10))
plot_tree(entropy_tree, feature_names=data.feature_names, class_names=data.target_names, filled=True)
plt.title("Entropy Tree")
plt.show()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(venv) PS C:\Users\MANYA_KM\Desktop\aml2> pip install matplotlib
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: six, pyparsing, pillow, packaging, kiwisolver, fonttools, cycler, contourpy, python-dateutil, matplotlib
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1 kiwisolver-1.5.0 matplotlib-3.10.9 packaging-26.2 pillow-12.2.0 pyparsing-3.3.2
python-dateutil-2.9.0.post0 six-1.17.0
(venv) PS C:\Users\MANYA_KM\Desktop\aml2> python dt.py
CART Accuracy: 0.9035087719298246
Entropy Tree Accuracy: 0.9122807017543859
Random Forest Accuracy: 0.9473684210526315
Extra Trees Accuracy: 0.9473684210526315
Gradient Boosting Accuracy: 0.9298245614035088
AdaBoost Accuracy: 0.9298245614035088
```

Ln 112, Col 32 Spaces: 4 UTF-8 CRLF { } Python 3.14.0 (venv)